Rigorous boundary-rank bounds

Exact certificates, formal separator theorem, and the remaining family gap

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Outcome

For a width-w Boolean boundary tensor flattened into $M \in \mathbb{Z}^{2^a \times 2^{w-a}}$, the following statements are now formalized in Lean and the finite lower bound has an independent native certificate:

$$0 \le \operatorname{rank}_{\mathbb{Q}}(M) \le \min(2^a, 2^{w-a}).$$

A nonzero determinant of a $k \times k$ minor proves $k \le \operatorname{rank}_{\mathbb{Q}}(M)$. If an exact linear separator factors $M = LR$ through d coordinates, then $\operatorname{rank}_{\mathbb{Q}}(M) \le d$. Therefore the certificate proves $k \le d$.

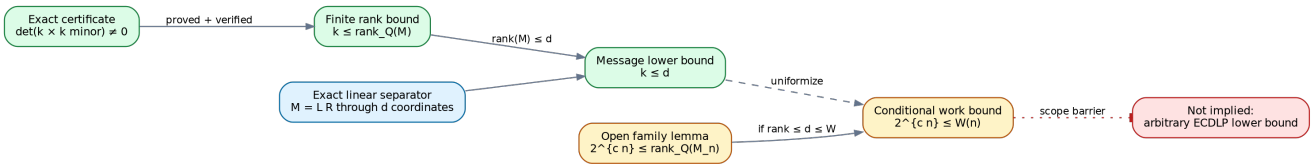


Figure 1: The proved finite implication, conditional asymptotic implication, and scope barrier. Green is established; amber requires the open family lemma; red is outside the method.

Certificate and verification

`tensor-census` schema v2 emits the split positions, selected rows and columns, and exact signed determinant for both the canonical and selected best balanced split. The verifier reconstructs the matrix from the exact boundary tensor and recomputes only that determinant using fraction-free Bareiss elimination. It does not call the rank routine.

The deterministic control checks 1,152 integer matrices from 1×1 through 6×6: exact fraction-free rank agrees with rational Gaussian elimination, and each discovered rank-profile minor has nonzero determinant. A changed determinant is rejected.

Formal theorem and family gap

The Lean project proves:

1. matrix and Boolean-flattening upper bounds;
2. nonzero minor ⇒ rank lower bound;
3. exact linear factorization ⇒ rank at most separator dimension;
4. their combined finite separator lower bound; and
5. if $2^{e(n)} \le \operatorname{rank}(M_n) \le d(n) \le W(n)$ and $cn \le e(n)$, then $2^{cn} \le W(n)$.

The last theorem is conditional by design. No uniform exponential minor or rank lower bound is currently proved for the production hashed prime-field family. No production semantic cut or E1 rank row has been generated.

Claim boundary

These results cover exact linear separator messages for the registered boundary matrices. They do not cover nonlinear compression, randomized or approximate contraction, alternate encodings, separator-avoiding algorithms, or arbitrary classical/quantum ECDLP algorithms. Even a future full-rank family theorem would refute this architecture, not subexponential ECDLP in general.

Reproducibility

The formal project pins Lean 4.35.0 and exact `crypto-lean` and Mathlib revisions. Full digests, statements, proof map, source citations, validation receipts, and the named open obstruction are in `RIGOROUS_BOUNDS.md`.